



# Global Migration Analysis: Understanding Through Economic & Demographic Lenses

Data Analytics Project

Presented by:

Suwash Chandra Gautam

Mouhssine Benaali

Grisham Tiwari

Presented to:

Prof. Dr. Nor Azizah Hitam

DS25 - Data Analytics

University of Europe for Applied Sciences

July 19, 2025

# Global Migration Analysis: Understanding Through Economic and Demographic Lenses

## Executive Summary

Migration is a deeply human decision — driven by opportunity, pressure, crisis, and ambition. This project explores the economic and demographic factors that drive net migration, using rich World Bank datasets spanning over six decades and covering more than 200 countries. In this project, we explored how GDP per capita, population size, and regional/income contexts influence net migration flows over time. Our goal is to uncover patterns, identify migration archetypes, and predict migration dynamics using machine learning. Using World Bank data for 1960–2023, we cleaned and merged historical datasets to build a comprehensive country-year panel, computed per-capita migration rates, and applied both statistical and machine learning methods to reveal key drivers and profiles of migration.

## Project Motivation and Goals

Core Question: Why do some countries consistently attract people, while others lose them? How do economic and demographic indicators explain this?

We sought to:

1. Analyze net migration trends over time.
2. Investigate economic and demographic correlations.
3. Cluster countries into migration archetypes.
4. Build predictive models to quantify the importance of features.
5. Compare inflows vs outflows across decades, regions, and income levels.

## Step 1: Data Loading, Cleaning, and Merging

Create a unified, analyzable dataset combining migration, population, and economic indicators from trusted World Bank sources.

1. Net Migration (1960–2023) – number of people entering or leaving each country.
2. Population – total population per country per year.
3. GDP per Capita (USD) – a proxy for economic opportunity and prosperity.
4. Region & Income Group Metadata

**Data Wrangling and Integration Process:**

All data preparation was performed using Python within a Jupyter Notebook environment, which allowed for iterative exploration, visualization, and transformation of the datasets. The key data wrangling steps included:

| Task                     | Description                                                              |
|--------------------------|--------------------------------------------------------------------------|
| Loaded CSVs              | Read Net Migration, Population, and GDP/capita into Pandas dataframes    |
| Reshaped Data            | Converted wide-format data into long-format using <code>pd.melt()</code> |
| Cleaned Country Names    | Standardized inconsistent naming (e.g., “United States” vs “US”)         |
| Handled Missing Values   | Dropped rows with incomplete migration or GDP data                       |
| Merged Datasets          | Joined all three datasets on Country Code and Year                       |
| Added Metadata           | Merged with World Bank Region & Income Group classification              |
| <i>Calculated Metric</i> | <i>Created Migration Rate per 1000 people for normalized comparison</i>  |

**Final Dataset Structure:**

After cleaning and merging, our unified dataframe includes:

| Column                  | Description                                                |
|-------------------------|------------------------------------------------------------|
| Country Name            | Name of the country                                        |
| Country Code            | ISO Alpha-3 code (eg. UAE, MYS, NPL, etc.)                 |
| Year                    | Year (1960–2023)                                           |
| Net_Migration           | Raw count of net migrants                                  |
| Population              | Total population                                           |
| GDP_per_Capita          | Country’s GDP per capita (in current USD)                  |
| Region                  | World Bank-defined region                                  |
| IncomeGroup             | World Bank income classification                           |
| Migration_Rate_per_1000 | Derived: $\text{Net Migration} / \text{Population} * 1000$ |

Result: We built a consistent panel dataset with around 5,000 country-year records from 1960 to 2023 — ready for analysis.

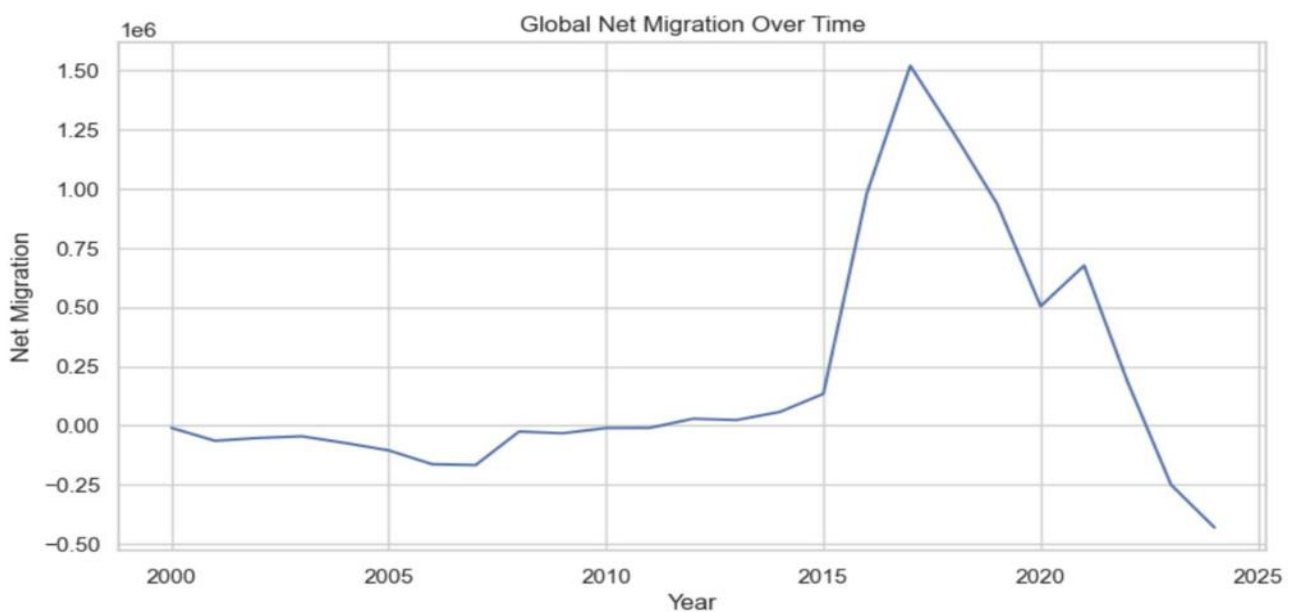
## Step 2: Exploratory Data Analysis (EDA)

### Time Frame Selection Rationale

- All indicators are available and consistent from 2000 to 2023.
- Allows for a reliable and unified analysis window.
- Captures recent global trends (e.g., economic shifts, refugee crises, etc.)

### A. Global Migration Trends Over Time

We aggregated net migration by year across all countries and visualized the global trend from 2000 to 2023.



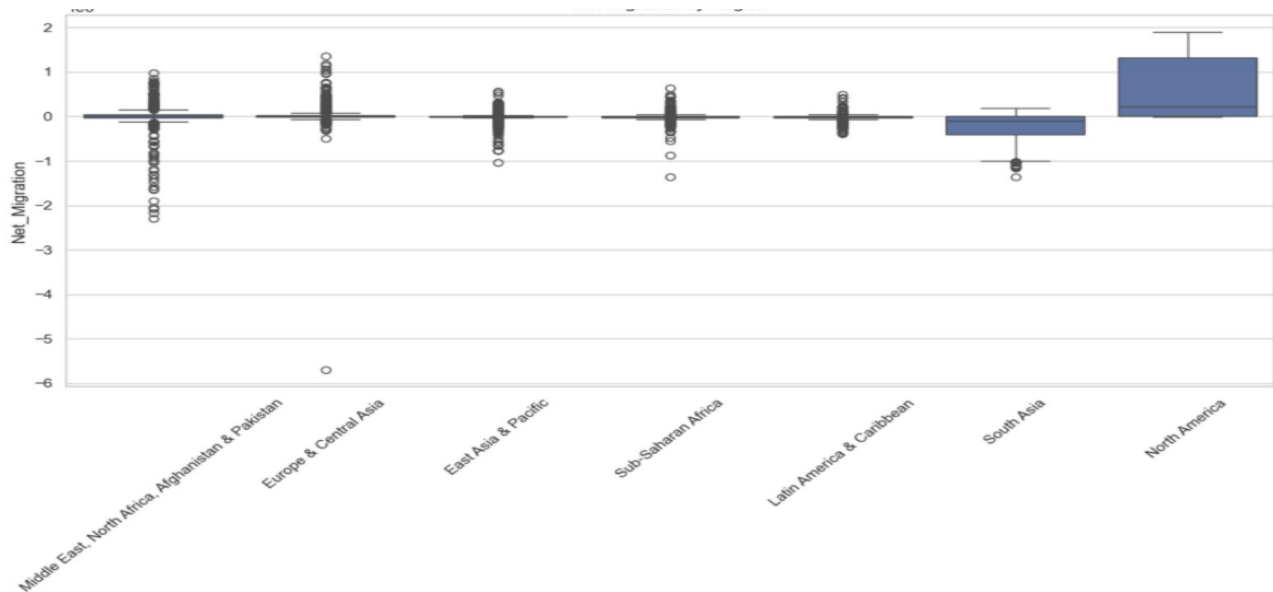
**Fig: Global Net Migration Over Time**

The world experienced fluctuating migration waves, with major peaks after 2015 and sharp decline in 2020 corresponding to global events such as:

- Conflicts and refugee crises in the Middle East and Africa
- EU expansion and globalization.
- COVID pandemic in 2020

### B. Migration by Region

We visualized regional migration distributions using boxplots to check how is the migration pattern in the region basis.



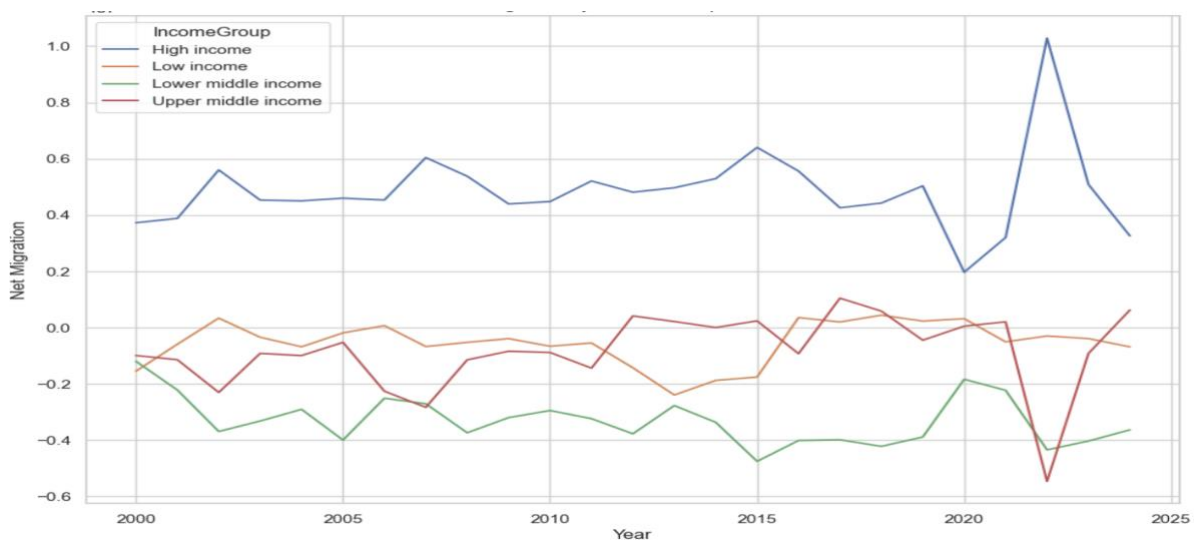
**Fig: Net Migration by Region**

Insight Highlights:

- North America consistently had net inflows.
- Sub-Saharan Africa and South Asia showed persistent net outflows.
- Middle East & North Africa showed sharp variation, driven by Gulf economies.

### C. Migration by Income Group

We grouped countries by World Bank income tiers (Low, Lower-Middle, Upper-Middle, High) and plotted their migration patterns over time.



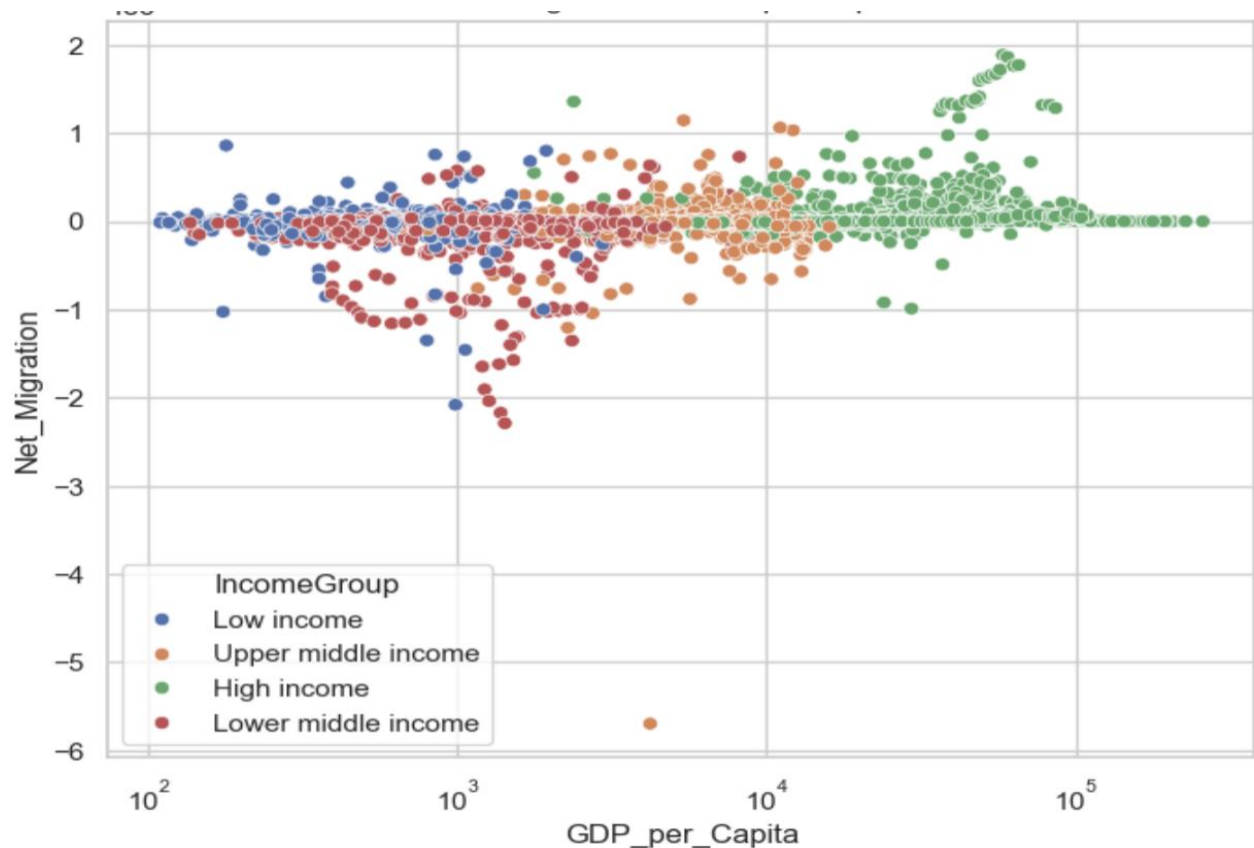
**Fig: Net Migration by Income Group Over Time**

## Key Findings:

- High-income countries saw the largest and most sustained inflows.
- Low-income countries had consistent outflows — likely due to economic push factors.
- Upper-middle-income countries fluctuated, showing both sending and receiving behaviors.

**D. Does GDP per Capita correlate with migration?**

We plotted Net Migration vs GDP per Capita on a log scale, colored by income group.



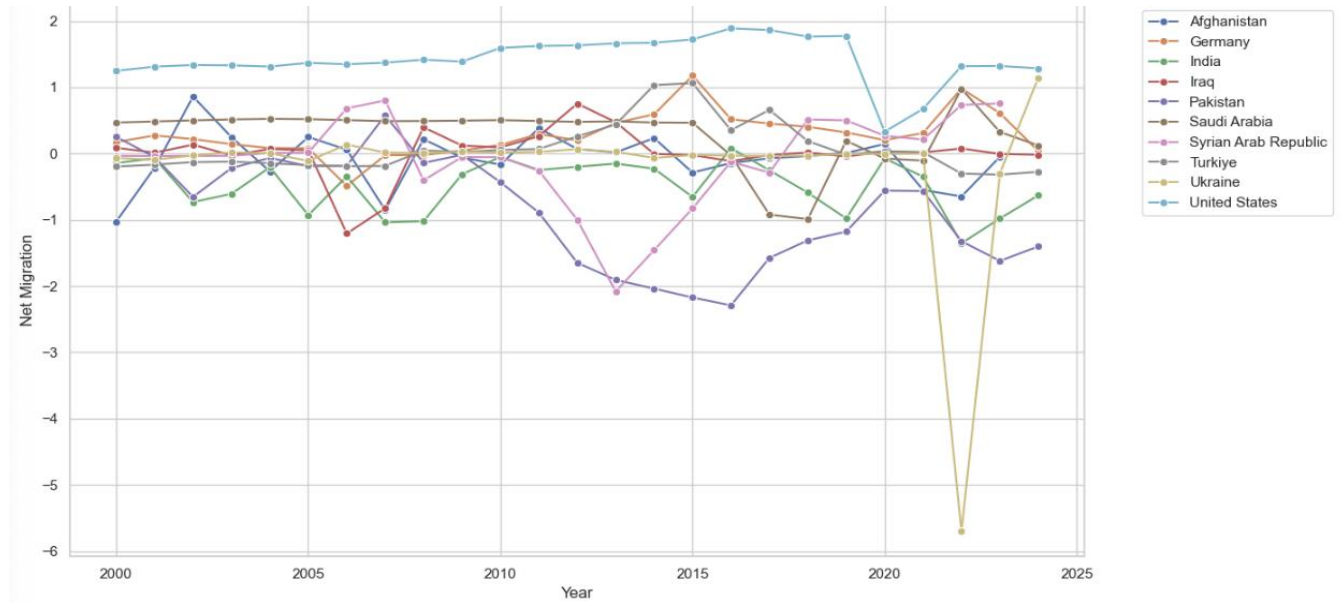
*Fig: Net Migration vs GDP per Capita*

## Observations:

- A positive correlation is visible: richer countries tend to have higher net migration.
- However, some low-income countries also show inflows — often due to labor migration or data anomalies.
- Outliers like Qatar, UAE, and Equatorial Guinea stand out for extremely high migration despite smaller populations.

### E. Migration Rate Volatility

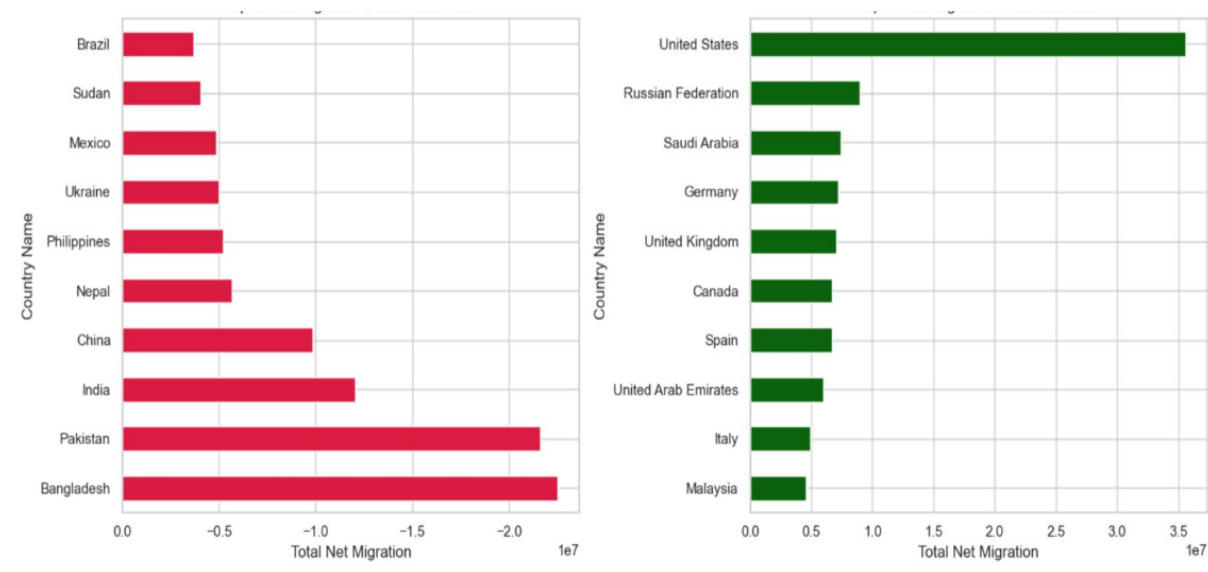
We measured the standard deviation of migration for each country and identified the top 10 most volatile countries.



**Fig: Net Migration Trends of Top 10 Most Volatile Countries**

Ukraine, Syria, Afghanistan, Iraq showed extreme swings — often due to war, climate, or economic collapse.

### F. Top Net Migration Inflow and Outflow Countries (Absolute Volume)



**Fig: Total Net Migration Inflow and Outflow Countries**

**Outflow Leaders:**

Countries like Bangladesh, Pakistan, India, and China experienced the largest total emigration, with net losses of over 20 million people in some cases. These outflows are largely driven by population pressure, economic hardship, and labor migration.

**Inflow Leaders:**

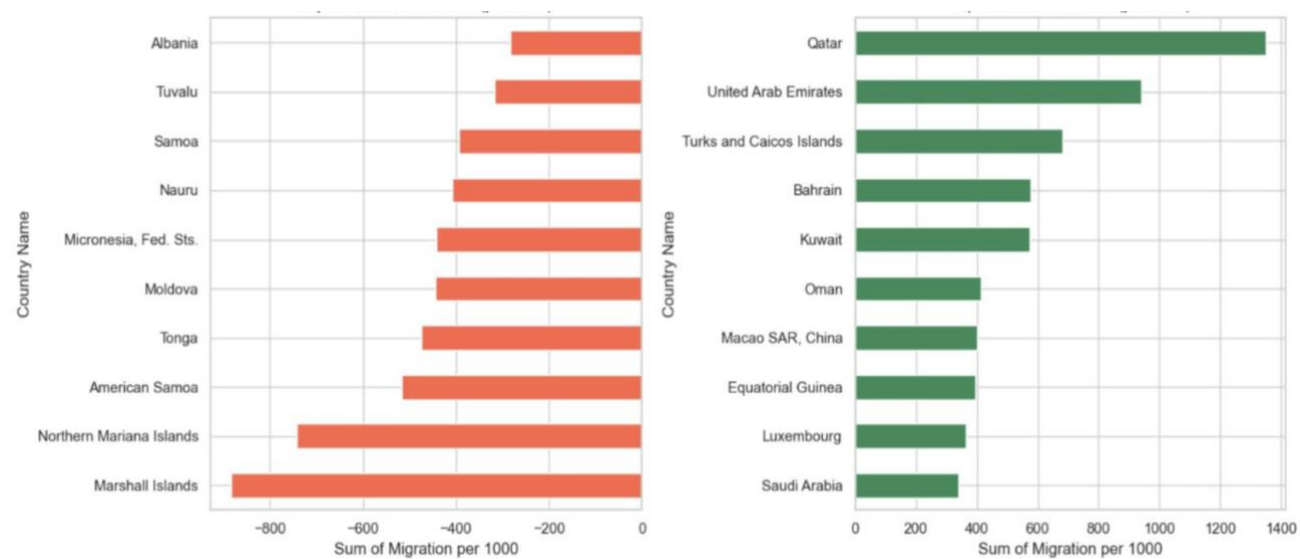
The United States tops the global chart with over 35 million net migrants, reflecting its role as the primary global destination for migrants. Other major inflow countries include Germany, UK, Canada, and Saudi Arabia – each benefiting from economic opportunities that attract foreign workers.

**Key Insight:**

This analysis shows that migration is not evenly distributed—some countries act as global migration magnets, while others serve as major sources of emigration. The flow patterns reflect disparities in economic opportunity, stability, and quality of life.

**G. Migration Extremes: Top Inflows and Outflows per 1000 People**

To understand which countries, experience the most extreme migration trends relative to their population size, we calculated the cumulative migration rate per 1,000 people across the full dataset period. This helps normalize migration by population and reveal hidden outliers that would otherwise be masked in absolute numbers.



*Fig: Top Inflows and Outflows per 1000 People*

### Top Outflow Countries per 1000 people (Left Panel)

The countries with the highest net emigration per capita are primarily:

- Small island nations (e.g., *Marshall Islands, Nauru, Micronesia, Tuvalu*)
- Remote or economically challenged regions (e.g., *Moldova, Albania, American Samoa*)

These countries consistently lose people due to limited economic opportunities, climate vulnerability, or isolation. Many rely on remittances and face long-term risks of population decline. The Marshall Islands shows a strikingly high cumulative outflow, with more than 800 emigrants per 1,000 residents over time—a powerful signal of migration as a survival strategy.

### Top Inflow Countries per 1000 people (Right Panel)

On the opposite end, countries with the highest net immigration per capita include:

- Oil-rich Gulf States (e.g., *Qatar, UAE, Bahrain, Kuwait, Saudi Arabia*)
- High-income microstates or tax havens (e.g., *Luxembourg, Macao SAR*)

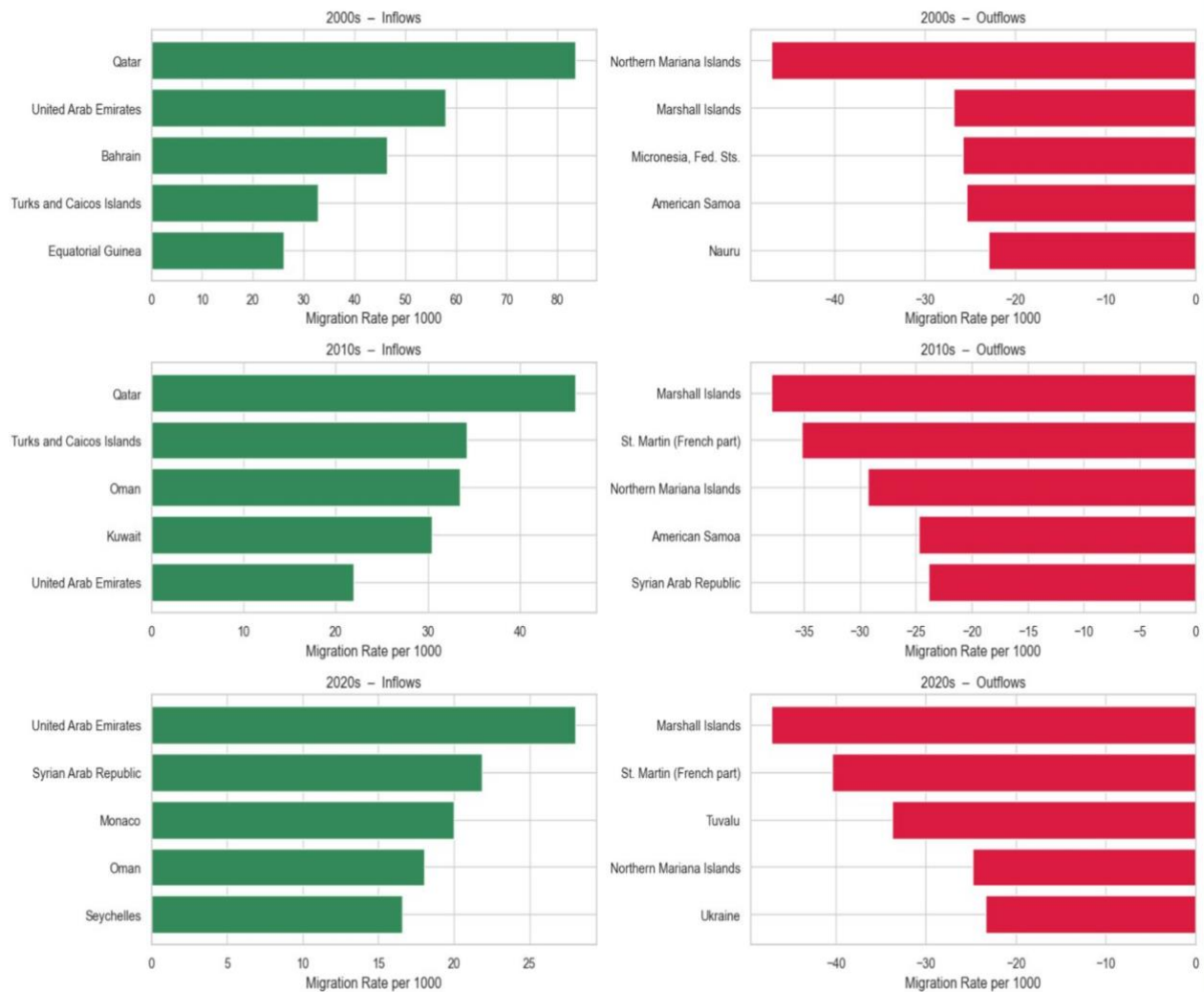
These countries attract large numbers of migrants relative to their population due to labor demand, high wages, and lenient immigration for foreign workers. Qatar tops the list with a staggering cumulative inflow exceeding 1,200 per 1,000 people—highlighting the scale of its dependency on foreign labor.

## H. Top Migration Winners and Losers by Decade

Using average migration rate per 1000 people, we ranked countries by:

- Top 5 net inflow countries
- Top 5 net outflow countries

For the 2000s, 2010s, and 2020s, countries like Qatar, UAE, and Oman repeatedly ranked as migration magnets. Conversely, small island nations and war-torn countries dominated the outflow list.



**Fig: Top Migration Winners and Losers by Decade**

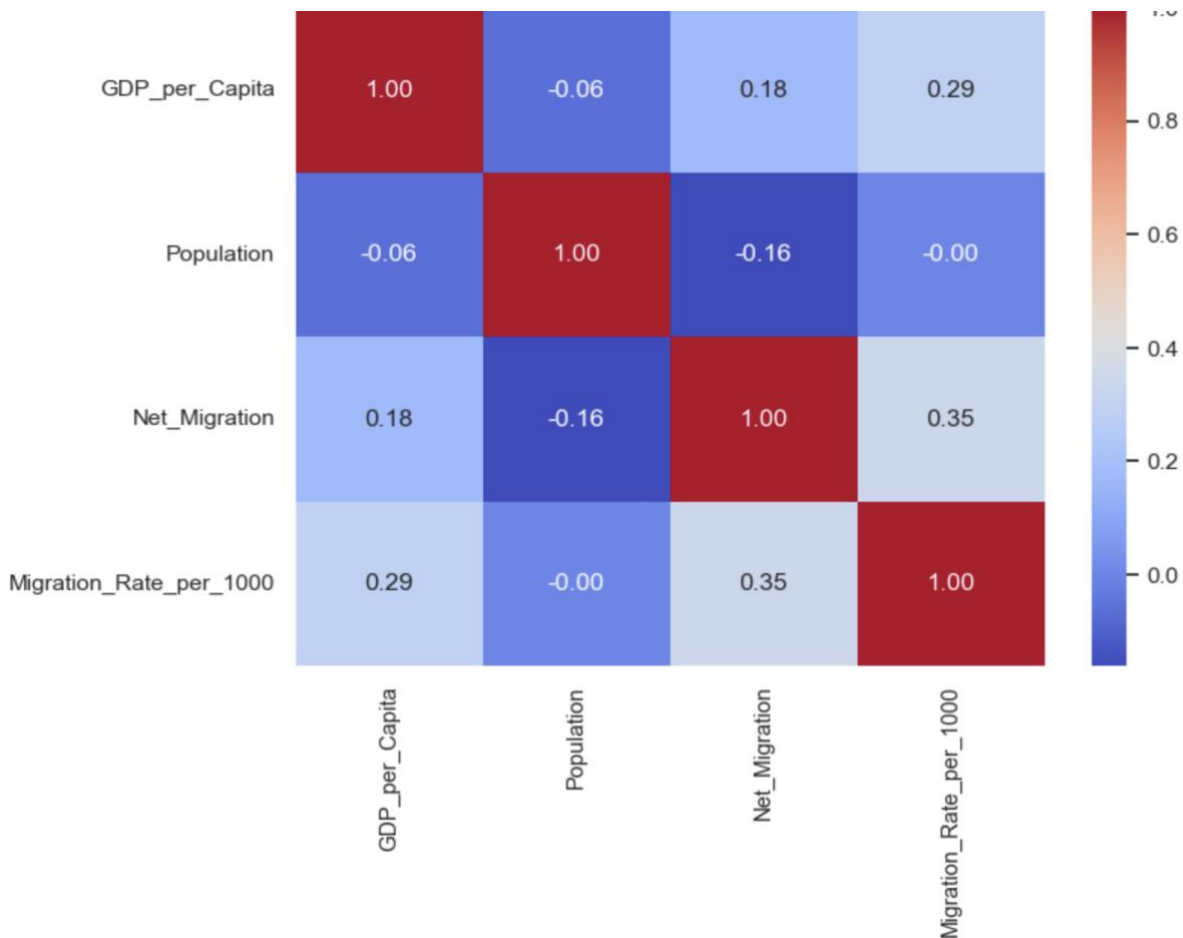
### Why We Measured Migration Two Ways: Volume vs. Impact

To gain a comprehensive understanding of migration dynamics, it is important to evaluate both net migration and migration per 1000 population. While net migration reflects the total volume of people entering or leaving a country—helpful for identifying global migration hubs—it tends to be dominated by larger countries due to their population size. In contrast, migration per 1000 population offers a normalized perspective, allowing us to assess the relative demographic impact of migration across nations of all sizes. This metric is especially insightful for highlighting significant shifts in smaller countries, where even modest changes in migration numbers can have a profound effect. Therefore, including both metrics enables a balanced analysis: one that captures the scale of migration and another that reveals its intensity and demographic significance.

## I. Correlation Matrix

We generated a heatmap of correlations between:

- GDP per capita
- Population
- Net Migration
- Migration Rate per 1000



*Visual: Correlation Heatmap*

Takeaway:

- Migration rate has moderate positive correlation with GDP per capita, and negative or weak ties with population.

### EDA Conclusion:

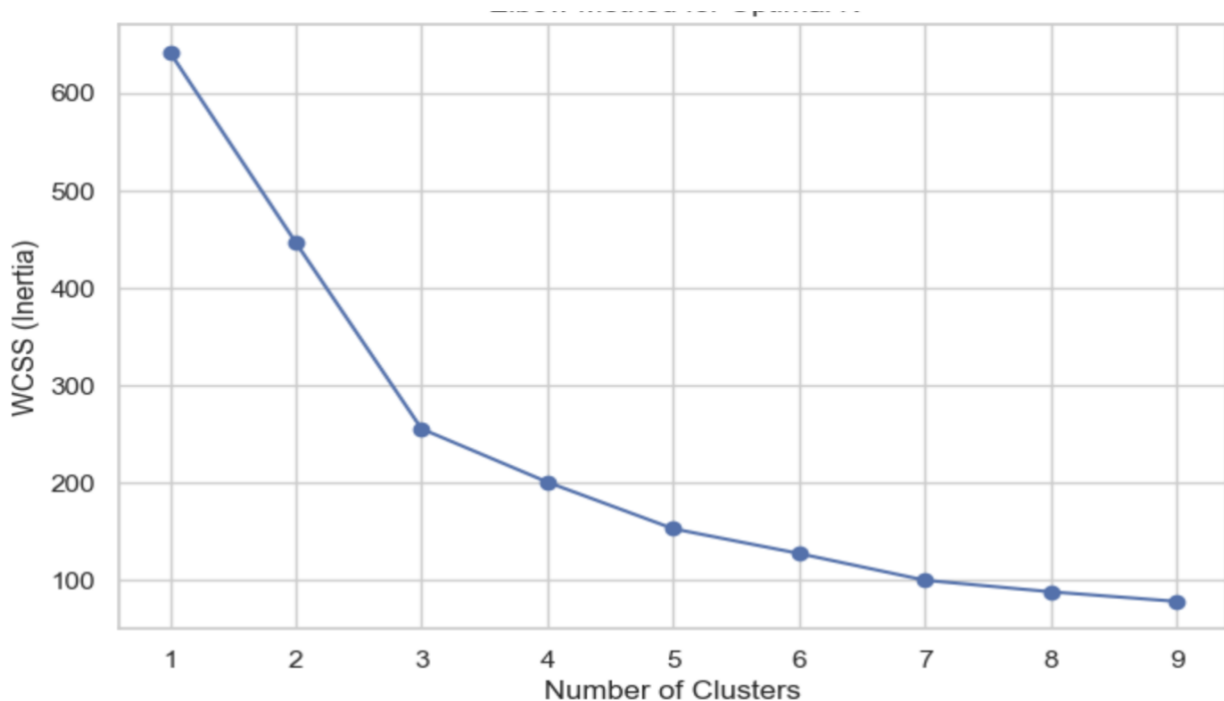
The data supports our hypothesis: Economic prosperity and population dynamics shape migration outcomes, but there are important exceptions and volatile patterns that require deeper modeling.

### Step 3: Clustering Countries by Migration Profile

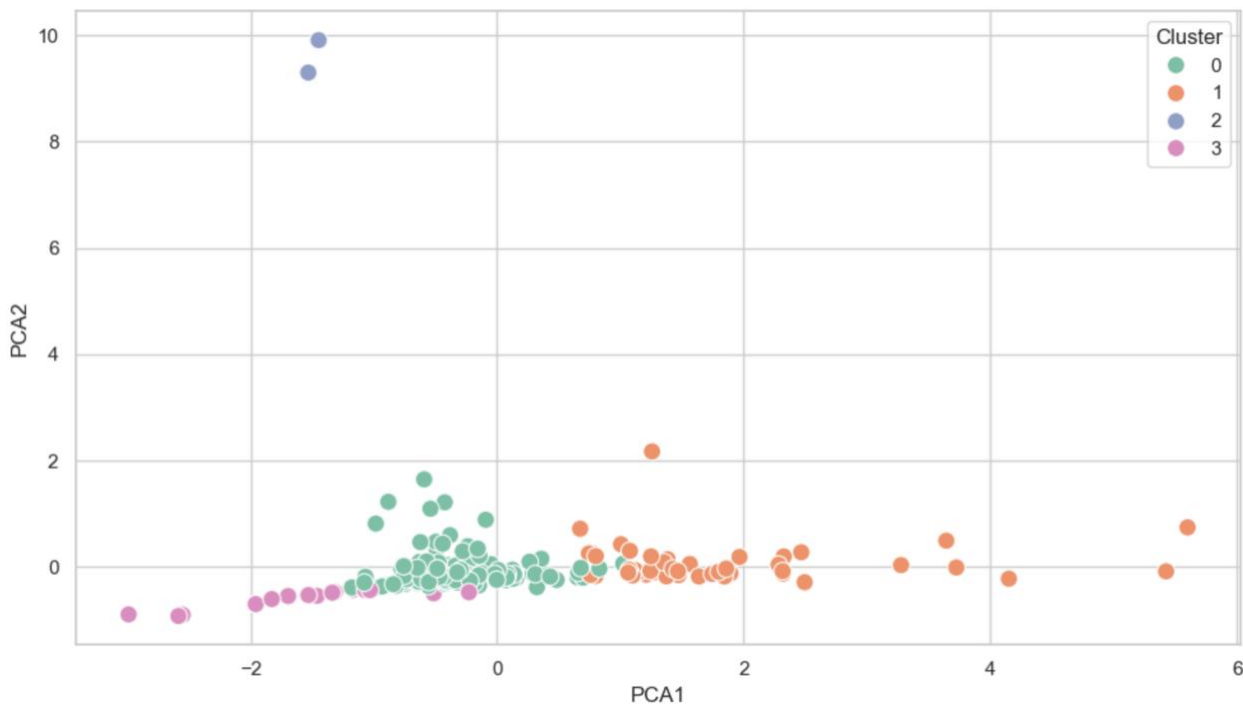
While visual EDA gave us broad regional and income-level trends, we wanted to discover hidden patterns by grouping countries based on Migration Rate per 1000, GDP per Capita and Population Size.

#### Methodology

- Features Used:
  - GDP per Capita
  - Population
  - Migration Rate per 1000 people
- Data Prep: Features were normalized using z-score scaling for fair comparison.
- Algorithm: We used K-Means Clustering, testing different values of k and choosing the one that best revealed clear distinctions (k=4).



*Visual: Elbow method for Optimal K*



*Visual: Country Clusters based on Migration, GDP and Population*

### Cluster Results Overview

| Cluster | Avg. GDP/capita | Avg. Pop.  | Avg. Mig. Rate | Label            |
|---------|-----------------|------------|----------------|------------------|
| 0       | Medium          | Mid-size   | Slightly -ve   | Sender States    |
| 1       | High            | Medium     | Strongly +ve   | Magnet Economies |
| 2       | Low             | Very Large | Mixed          | Swing Giants     |
| 3       | High            | Small      | Very -ve       | Wealthy Leavers  |

### Cluster Interpretations:

#### Cluster 0 – Middle-Income Migrant Balancers:

- Profile: These countries have moderate GDP per capita, large populations, and near-zero or slightly negative migration rates.
- Typical Dynamics: They often face dual pressures — sending and receiving migrants depending on economic cycles or political context.
- Example Countries:
  - Afghanistan, Algeria, Angola, Antigua and Barbuda, Argentina, Armenia, Aruba, Azerbaijan, Bahamas, Bangladesh
- Insight: These are developing or emerging nations with transitional migration behaviour — neither strong attractors nor major senders.

**Cluster 1 – High-Income Magnets:**

- Profile: Countries with high GDP per capita, moderate population sizes, and strong positive migration rates.
- Typical Dynamics: Attractive for labour migration, higher education, and quality of life.
- Example Countries:
  - Andorra, Australia, Austria, Bahrain, Belgium, Bermuda, Canada, Cayman Islands, Channel Islands, Cyprus
- Insight: These are advanced economies consistently drawing immigrants due to economic opportunity and stability.

**Cluster 2 – Populous Powerhouses:**

- Profile: Extremely large populations, moderate GDP per capita, and near-zero net migration rates.
- Example Countries:
  - China, India
- Insight: Despite their size and global influence, net migration is balanced, likely due to internal labour market absorption and controlled emigration.

**Cluster 3 – Small-State Leavers:**

- Profile: Small population countries with low to moderate GDP per capita, and significantly negative migration rates.
- Typical Dynamics: These nations are experiencing strong outmigration due to limited opportunities.
- Example Countries:
  - Albania, American Samoa, Guam, Guyana, Kosovo, Marshall Islands, Micronesia, Moldova, Nauru, Northern Mariana Islands
- Insight: These countries often suffer from "brain drain" and economic challenges, driving residents to seek better lives abroad.

**Clustering Conclusion:**

Our clustering revealed distinct migration profiles that cut across traditional region/income lines. These insights help policymakers and researchers frame migration patterns in typologies rather than only geopolitics.

## Step 4: Feature Importance Analysis

After uncovering distinct migration profiles through clustering, we turned to a key question: What drives migration outcomes—and how much does each factor matter? To answer this, we applied a Random Forest Regressor to estimate how strongly each variable contributes to a country's net migration rate per 1,000 people.

### Methodology:

We prepared the data by:

- Dropping missing values in key variables.
- One-hot encoding the categorical columns Region and Income Group to include them as model features.
- Excluding identifier and target columns such as Country Name, Country Code, Year, and Net\_Migration.

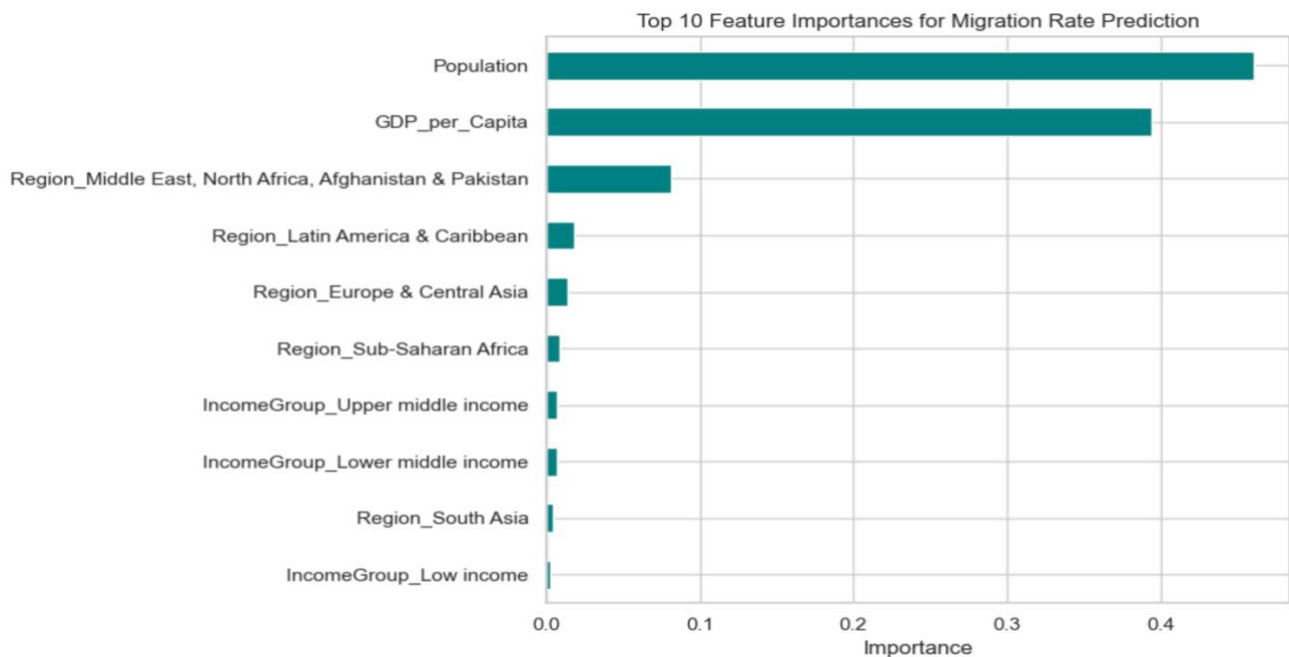
Then:

- Split the dataset into training (80%) and testing (20%) subsets.
- Trained a RandomForestRegressor with 100 estimators.
- Evaluated model performance using  $R^2$  Score, achieving:

***$R^2$  Score: 0.3921***

This suggests that nearly 39% of the variance in migration rate can be explained by our chosen features. The model also allowed us to quantify feature importance, revealing which variables had the greatest impact on migration patterns.

| Rank | Features                             | Description           |
|------|--------------------------------------|-----------------------|
| 1    | Population                           | Demographic pressure  |
| 2    | GDP_per_Capita                       | Economic pull factor  |
| 3    | Region_Middle East, North Africa.... | Regional influence    |
| 4    | Region_Latin America & Caribbean     | Regional influence    |
| 5    | Region_Europe & Central Asia         | Regional influence    |
| 6    | Region_Sub-Saharan Africa            | Regional influence    |
| 7    | IncomeGroup_Upper middle income      | Income-level dynamics |
| 8    | IncomeGroup_Lower middle income      | Income-level dynamics |
| 9    | Region_South Asia                    | Regional influence    |
| 10   | Incomegroup_Low income               | Income-level dynamics |



*Visual: Feature Importance Plot*

### Insight:

GDP per capita and population are not just intuitive factors—they're empirically powerful in shaping migration trends. The prominence of regional dummies further underscores the contextual nature of migration.

### Step 5: Predictive Modeling

Objective: To understand what factors, influence net migration rates per 1000 people, and how well we can predict migration behavior using economic and demographic variables.

### Modeling Setup:

Target Variable:

- Migration\_Rate\_per\_1000

Input Features:

- GDP\_per\_Capita
- Population

(We kept the model simple and interpretable, using only strong numerical predictors.)

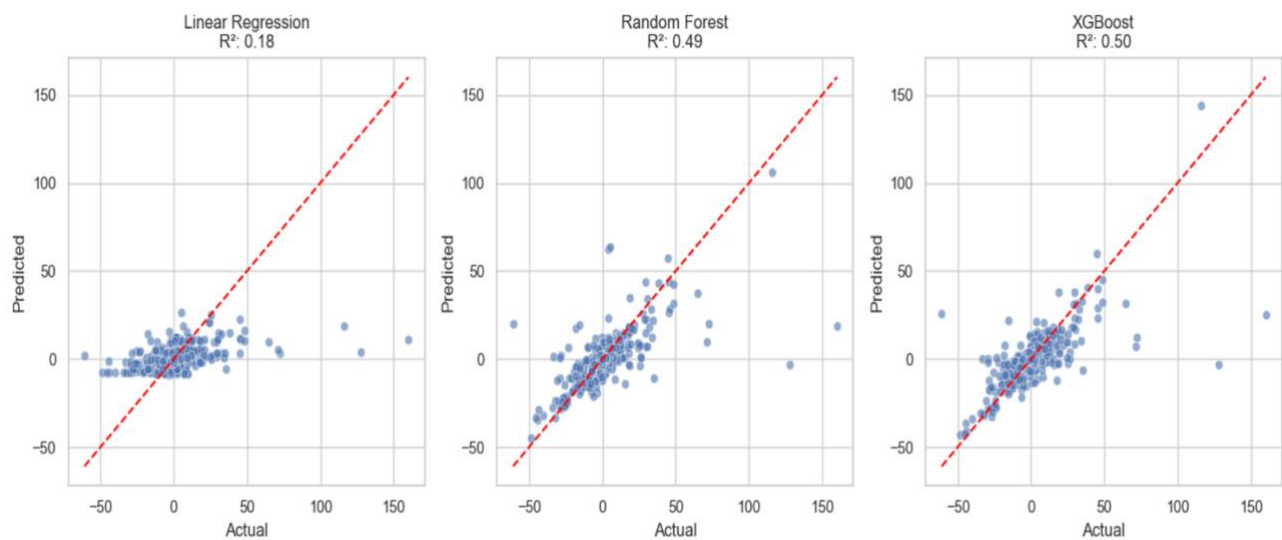
## Models Trained:

We trained and compared three models:

1. Linear Regression
2. Random Forest Regressor
3. XGBoost Regressor

All models were evaluated on:

- $R^2$  Score (how much variance is explained)
- MAE (Mean Absolute Error – prediction error magnitude)



*Visual: Various Models Comparison*

## Model Comparison Results:

| Model             | $R^2$ Score | RMSE  | Comment                                                                         |
|-------------------|-------------|-------|---------------------------------------------------------------------------------|
| Linear Regression | 0.179       | 11.82 | Very basic model. Explains only ~18% of the variation. High error.              |
| Random Forest     | 0.490       | 9.32  | Performs much better. Captures nonlinear relationships.                         |
| XGBoost           | 0.498       | 9.25  | Best performer here. Slightly better than Random Forest in both $R^2$ and RMSE. |

**Interpretation:**

- Linear Regression performs the weakest, with an  $R^2$  of 0.179, meaning it explains only 18% of the variation in migration. The relatively high RMSE (~11.82) indicates large prediction errors.
- Random Forest and XGBoost both show significantly better performance, explaining 50% of the variation in migration. They also reduce the error by about 20% compared to linear regression.
- XGBoost edges out slightly as the best model, with the lowest RMSE (9.25) and highest  $R^2$  (0.498), though only marginally better than Random Forest.

The input variables likely GDP per capita, population, explain about 50% of the variation in migration patterns. This means economic and demographic variables have a moderate influence on migration trends. Other considerable factors likely include conflict/violence, climate conditions, border openness, language/diaspora presence, educational opportunities, political stability etc.

**Key Insights & Takeaways:**

- Migration is shaped by both push and pull: Poor economic conditions and population pressure push people out; strong economies and regional appeal pull people in.
- Tiny island states and fragile regions face high volatility and sustained losses.
- High-income, service-driven countries (e.g., UAE, Qatar) attract large inflows relative to size.
- Economic indicators alone don't tell the full story—regional and income context matters.

**What This Means for Policy & Research**

- Policy makers in sender countries can use these insights to address root causes like poverty and lack of opportunity.
- Receiver countries can better anticipate migratory pressures and plan infrastructure accordingly.
- Further research could include refugee data, climate variables, human development index, or governance metrics to enrich these findings.

**References**

- World Bank (2024). *World Development Indicators*. Retrieved from <http://data.worldbank.org>
- United Nations DESA (2022). *International Migration 2020 Highlights*. Population Division, Department of Economic and Social Affairs.

**AI Assistance Acknowledgment:** This report has been polished using AI tools to enhance clarity, structure, and readability.